

1 Regularisation in Machine Learning:

↳ Regularisation is one of the most important concepts of machine learning.

↳ It is a technique to prevent the model from overfitting by adding extra information to it.

↳ Sometimes the machine learning model performs well with the training data but does not perform well with the test data.

↳ Thus the model is not able to predict the output when deals with unseen data by introducing noise in the output and hence the model is called overfitted. This problem can be dealt with the help of a regularization technique.

↳ ~~This~~ This

↳ This technique can be used in such a way that it will allow to maintain all variables or features in the model by reducing the magnitude of variables.

Techniques of Regularization

There are mainly two types of regularization techniques, which are given below

- Ridge Regression

- Lasso Regression

Ridge Regression

↳ Ridge regression is one of the types of linear regression in which a small amount of bias is introduced so that we can get better long-term prediction.

↳ Ridge regression is a regularization technique, which is used to reduce the complexity of the model. It is also called as L2 regularization.

↳ In the technique, the cost function is altered by adding to ~~reduce~~ the complexity the penalty term to it.

↳ The amount of bias added to model is called Ridge Regression Penalty. We can calculate it by multiplying with the lambda to the squared weight of each individual ~~for~~ feature.

↳ The equation for the cost function in ridge regression will be:

$$\sum_{i=1}^M (y_i - \hat{y}_i)^2 + \lambda \sum_{j=0}^n \beta_j^2$$

↳ In the above equation, the penalty term regularizes the coefficients of the model and hence ridge regression reduces the amplitude of the coefficients that decreases the complexity of the model.

↳ As we can see from the above equation, if the values of λ tend to zero, the equation becomes the cost function of the linear regression model. Hence, for the minimum value of λ , the model will resemble the linear regression model.

↳ A general linear or polynomial regression will fail if there is high collinearity b/w the independent variables, so to solve such problem, Ridge regression can be used.

↳ It helps to solve the problem if we have more parameters than samples.

Ridge Regression

- ↳ It is a regularized version of linear regression that adds a penalty term to the cost function.
- ↳ It is mainly used to solve the problem of multicollinearity (high correlation among independent variables) and to prevent overfitting.
- ↳ It is supervised machine learning algo used for predicting continuous values.
- ↳ It adds L2 penalty on regression coefficients.

Role of Regularization parameter (λ)

λ value	effect
1) $\lambda = 0$	Same as Linear Regression
2) Small λ	Slight shrinkage of coefficients
3) Moderate λ	Better balance b/w bias & variance
4) Large λ	Coefficient become very small
5) Very large λ	Model may underfit

Why Ridge Regression?

- Feature are highly correlated
- Number of predictors is large
- Model overfits training data

Objective Function & Mathematical Formulation

↳ Ridge regression adds L_2 penalty the

OLS loss function:
(Ordinary least squares)

OLS, the objective is to minimize the Residual sum squares (RSS)

$$RSS = \sum_{i=1}^n \left(y_i - \beta_0 - \sum_{j=1}^p \beta_j x_{ij} \right)^2$$

p = no of features

n = no. of sample

Thus objective function

$$J(\beta) = \sum_{i=1}^n \left(y - \sum_{j=1}^p x_{ij} \beta_j \right)^2 + \lambda \sum_{j=1}^p \beta_j^2$$

In matrix notation

$$J(\beta) = (y - X\beta)^T (y - X\beta) + \lambda \|\beta\|^2$$

(1) $\lambda = 0$; Ridge regression become linear regression

$\lambda \rightarrow \infty$; all feature weights $\beta_j \rightarrow 0$

Lasso Regression: (Least Absolute Shrinkage & Selection)

↳ It is a regression analysis method that performs both variable selection and regularization in order to enhance the prediction accuracy and interpretability of resulting statistical model.

↳ Lasso was originally formulated for linear regression models.

↳ It is also known as ℓ_1 regularization.

↳ It is used over regression method for a more accurate prediction.

↳ This model uses shrinkage. Shrinkage is where data values are shrunk towards a central point as ~~mean~~ ^{mean}.

↳ The lasso procedure encourages simple, sparse models (i.e. models with fewer parameters).

↳ This particular type of regression is well-suited for models showing high levels of multicollinearity or when you want to automate certain parts of model selection, like variable selection/parameter elimination.

Step-by-Step explanation of Lasso Regression

1. Linear Regression model:

LASSO regression model starts with the standard linear regression model, which assumes a linear relationship between the independent variables (features) and the dependent variable (target). The linear regression model equation can be represented as follows

$$y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \dots + \beta_p x_p + \epsilon$$

where y is dependent variable

- $\beta_0, \beta_1, \dots, \beta_p$ are coefficients (parameters) to be estimated.
- $x_1, x_2, \dots, x_p$ are the independent variables
- ϵ represents the error term.

2. L1 regularization:

LASSO regression introduces an additional penalty term based on the absolute values of the coefficients. The L1 regularization term is the sum of the absolute values of the coefficients multiplied by a tuning parameter λ :

$$L_1 = \lambda * (|\beta_1| + |\beta_2| + \dots + |\beta_p|) \text{ where}$$

- λ is regularization parameter that controls the amount of regularization applied.
- $\beta_1, \beta_2, \dots, \beta_p$ are the coefficients.

3. Objective Function:

The objective of LASSO regression is to find the value of the coefficients that minimize the sum of the squared difference b/w the predicted values and actual values. while also minimizing the L_1 regularization term:

minimize: $RSS + L_1$ where

• RSS is the residual sum of squares, which measures the error between the predicted values and actual values.

4. Shrinking Coefficients:

By adding the L_1 regularization term, LASSO regression can shrink the coefficients towards zero. when λ is sufficiently large, some coefficients are driven to exactly zero. This property of LASSO makes it useful for feature selection, as the variables with zero coefficients are effectively removed from the model.

5. Tuning Parameter (λ):

The choice of the regularization parameter λ is crucial in LASSO regression. A large λ value increases the amount of regularization, leading to more coefficients being pushed towards zero.

Conversely, a smaller λ value reduces the regularization effect, allowing more variables to have non-zero coefficients.

5. Model fitting:

To estimate the coefficient in LASSO regression, an optimisation algorithm is used to minimize the objective function. Coordinate descent is commonly employed, which iteratively updates each coefficient while holding the other fixed.

→ Thus LASSO regression offers a powerful framework - work for both prediction and feature selection, especially when dealing with high-dimensional datasets where the number of feature is large.

	Ridge Regression	Lasso Regression
Regularization	L_2 Penalty	L_1 Penalty
Feature Selection	Does not support feature selection	Doesn't support feature selection
Coefficient Shrinkage	Smooth shrinkage	Aggressive shrinkage
Zero Coefficient	Never	Possible
Multicollinearity	Handles very well	Moderate handling
Interpretability	Lower	Higher